SUSHMIT PRASHANT GUJAR

Senior AI/ML Engineer

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 8+ years of experience building production-grade machine learning, generative AI, RAG, agentic AI, and data engineering solutions across healthcare, telecom, financial crime, and regulated enterprise environments. Proven track record architecting scalable ML platforms, knowledge graphs, multi-agent systems, real-time inference pipelines, MLOps workflows, and cloud-native data products that reduce manual effort, improve decision accuracy, accelerate research, and convert complex enterprise data into measurable business impact.

SKILLS

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Agentic AI, multi-agent systems, prompt engineering, embeddings, semantic search, vector retrieval, synthetic data generation, LLM evaluation, grounding, hallucination mitigation, explainable AI (XAI)

Machine Learning & MLOps: Predictive modeling, anomaly detection, NLP, deep learning, feature engineering, model monitoring, CI/CD, MLflow, Kubeflow, Airflow, Docker, Kubernetes, model retraining, drift-aware feedback loops

Data Platforms & Cloud: Python, PySpark, SQL, Azure Databricks, Delta Lake, Snowflake, AWS Lambda, Dgraph, streaming pipelines, batch processing, data governance, data quality, scalable ETL/ELT architecture

PROFESSIONAL EXPERIENCE

Johnson & Johnson April 2025 – Present

AI/ML Engineer (Innovative Medicine) *Remote*

- **Architected enterprise knowledge graph foundation for XAI:** Led ontology design and production deployment of domain-specific knowledge graphs across supplier data, drug manufacturing, batch release, and stability study domains, establishing a scalable Dgraph backbone for explainable, governed AI use cases.
- **Pioneered agentic AI for regulatory intelligence:** Built autonomous document intelligence agents that navigate, extract, synthesize, and summarize quality and supplier documentation, accelerating knowledge discovery and reducing manual research effort for internal business users.
- **Delivered advanced analytics for quality issue resolution:** Developed ML models to classify manufacturing deviations and supplier incidents, cutting manual investigation time by 40% and improving incident identification accuracy by 15%.
- **Engineered secure multi-agent interoperability:** Designed an agent-to-agent communication protocol connecting internal AI agents with third-party vendor agents, enabling governed automation for complex enterprise data synthesis workflows.
- **Built RAG pipelines for GxP document intelligence:** Developed end-to-end retrieval pipelines for supplier qualifications, lab specifications, and drug product/substance documents, including indexing, chunking, retrieval, and context optimization for regulated knowledge access.
- **Applied generative AI for privacy-safe test data synthesis:** Integrated synthetic tabular data generation approaches for supply chain risk modeling and internal testing, improving development velocity while preserving sensitive enterprise data privacy.
- **Strengthened data integrity and governance:** Built PySpark and Azure Databricks pipelines to cleanse, standardize, and validate batch quality and supply chain logistics records, improving data accuracy and reducing manual review time by 40%.
- Led global advanced analytics delivery by directing distributed technical resources across India and partnering with stakeholders in Belgium to deliver AI and data solutions for GxP-regulated quality and supply chain operations.

Comcast January 2019 – April 2025

Scientist 3, Enterprise Business Intelligence *Philadelphia, PA*

- **LLM-based customer churn and retention modeling:** Designed, trained, and deployed a real-time LLM-powered retention platform analyzing live call and chat transcripts to generate personalized offers, increasing offer acceptance by 20% and supporting high-cardinality churn modeling with MLflow, Databricks, and feature store capabilities.
- **Feedback-driven MLOps loop:** Built a closed-loop retraining workflow on Databricks Workflows that incorporated real-time agent ratings into weekly fine-tuning cycles, improving recommendation relevance as customer behavior and frontline feedback evolved.
- **Real-time LLM inference for proactive issue resolution:** Architected streaming ML pipelines for call and chat telemetry, transforming sentiment, intent, and escalation signals into near-real-time routing decisions that reduced average handle time by 18% and lowered escalations by 30%.
- **LLM-driven sentiment and root-cause classification:** Productionized CI/CD workflows for sentiment and complaint categorization models, automating labeling, retraining, and deployment while achieving 95% accuracy in primary complaint-driver detection.
- **Cloud-native MLOps and infrastructure automation:** Developed Docker and Kubernetes-based deployment modules with blue/green release strategies, environment-specific configuration, A/B testing support, and monitoring for reliable customer-experience ML services.
- **Xfinity Mobile finance and data engineering:** Optimized PySpark and Databricks batch pipelines processing millions of daily billing, usage, and churn records, providing scalable data infrastructure for analytics, finance, and ML use cases.
- **Statistical modeling and feature engineering:** Built PySpark analytical pipelines for property penetration, customer adoption, and competitive intelligence, accelerating strategic Wi-Fi Ready expansion decisions.
- **Geospatial fiber footprint analytics:** Integrated geospatial and competitive datasets to identify fiber-overbuild exposure across markets, revealing that 40% of analyzed regions faced fiber competition.

Regulatory Data Corp March 2018 – January 2019

Data Engineer *King of Prussia, PA*

- Engineered real-time pipelines in Python, PySpark, and AWS Lambda to process high-velocity transaction streams and surface anomalous financial activity for fraud and compliance teams.
- **Unsupervised AML risk scoring:** Implemented anomaly detection models with Delta Lake and Kubeflow Pipelines for automated retraining, reducing false positives by 25%.
- **Cloud-native fraud detection MLOps:** Architected Dockerized microservices and CI/CD workflows for ensemble fraud models, integrating SHAP-based interpretability to support regulatory transparency and achieving a 28% reduction in false positives plus a 19% uplift in early fraud detection.
- **Deep learning NLP for suspicious transaction analysis:** Built entity recognition and sequence classification pipelines to extract risk signals from unstructured transaction descriptions, increasing identification of previously undetected suspicious entities by 23%.

Infosys June 2015 – August 2016

Data Engineer *Pune, India*

- Designed PySpark and Apache Airflow pipelines for mortgage risk assessment and fraud detection, reducing data processing time by 35%.
- Standardized loan application data across SQL, Snowflake, and Delta Lake environments, improving data integrity and reducing processing errors by 30%.
- Orchestrated end-to-end data workflows for regulatory reporting, automating report generation and reducing manual effort by 50%.
- Integrated credit bureau, KYC, transaction, and internal risk datasets to strengthen fraud modeling and improve detection accuracy by 20%.

EDUCATION

M.S., Data Science — Rutgers University, New Brunswick September 2016 – December 2017

Coursework: Machine Learning, Deep Learning, Data Science, Algorithms

B.E., Electronics Engineering — Mumbai University, Mumbai July 2011 – May 2015

Coursework: Java, C++, Embedded Systems, Object Oriented Programming, Microprocessors, Robotics